What the program does:

This program performs four main functions: encrypting messages using a user-specified shift value, decrypting messages using the corresponding shift value, runs a brute-force attack by testing shift values from 1 to 10, and storing and retrieving the history of encrypted and decrypted messages. It also allows the user to exit the program when desired.

How it works:

The program works by accepting user input through an input function offering five options. The five options are:

1. E – Encryption
2. D – Decryption
3. B – Brute Force Attack
4. M – Memory
5. 0 (zero) – Exit

Depending on the option chosen by the user, the program directs them to the corresponding functions. The encryption function prompts the user to enter a shift value, then uses Python's built in mathematical operations to shift each letter in the message by the specified number of positions in the alphabet, producing an encrypted message.

```
C:\Users\stacy\PycharmProjects\PythonProject1\.venv\Scripts\python.exe C:\Users\stacy\PycharmProjects\PythonProject1\Bentley.Forbes-ceaser_cipher.py
Enter your choice:
 E for Encryption
 D for Decryption
 B for Brute Force Attack
 M for Memory
 0 for Exit
E
How many encrypted messages do you choose to store? (1-20) 1
Enter the shift value: 5
Enter the message to encrypt: Assignment Completed
Encrypted message: Shift Value= 5, Encryption: Fxxnlsrjsy Htruqjyji
```

The decryption function uses the same code as encryption but reverses the process to recover the original message. The user must provide the shift value used during encryption to correctly decrypt the message. This adds a level of security, as anyone who obtains the encrypted message would need to know the shift value to reveal its contents.

```
Enter your choice:
 E for Encryption
 D for Decryption
 B for Brute Force Attack
 M for Memory
 0 for Exit
D
How many encrypted messages do you choose to store? (1-20) 1
Enter the shift value: 5
Enter the message to decrypt: Fxxnlsrjsy Htruqjyji
Decrypted message: Shift Value= 5, Decryption: Assignment Completed
```

The Brute Force Attack feature simulates a basic decryption attack by testing shift values from 1 to 10. It displays the resulting version of the encrypted message for each shift value, allowing the user to compare the outputs and identify the one that reveals the original message.

```
Decrypted message: Shift Value= 5, Decryption: Assignment Completed
Enter your choice:
 E for Encryption
 D for Decryption
 B for Brute Force Attack
 M for Memory
 0 for Exit
B
Enter the message to decrypt: Fxxnlsrjsy Htruqjyji
Shift value 1: Ewwmkrqirx Gsqtpixih
Shift value 2: Dvvljqphqw Frpsohwhg
Shift value 3: Cuukipogpv Eqorngvgf
Shift value 4: Bttjhonfou Dpnqmfufe
Shift value 5: Assignment Completed
Shift value 6: Zrrhfmldms Bnlokdsdc
Shift value 7: Yqqgelkclr Amknjcrcb
Shift value 8: Xppfdkjbkq Zljmibqba
Shift value 9: Wooecjiajp Ykilhapaz
Shift value 10: Vnndbihzio Xjhkgzozy
```

The Memory feature uses a dictionary to store encrypted messages along with their corresponding decrypted results during the session. It serves as a history of the messages processed by the program, allowing users to view previously stored information.

```
Enter your choice:
 E for Encryption
 D for Decryption
 B for Brute Force Attack
 M for Memory
 0 for Exit
M
{'Fxxnlsrjsy Htruqjyji': 'Shift Value= 5, Decryption: Assignment Completed'}
```

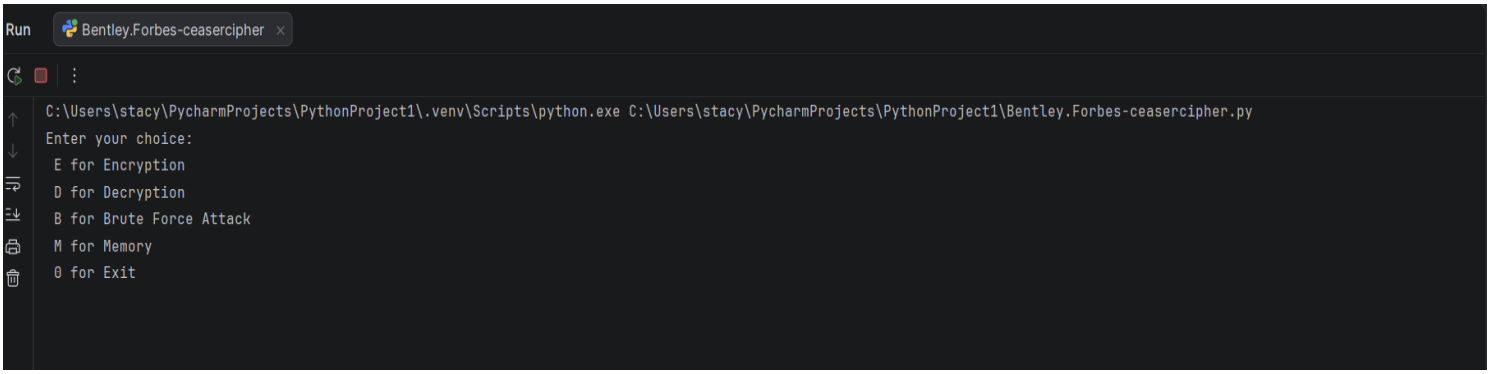
Zero “0” is used to exit the program, used because zero in binary means off. The program also handles incorrect inputs by terminating whenever the user fails to follow the on-screen instructions. This error-handling approach was intentionally chosen to frustrate anyone attempting to disrupt the program by entering invalid values, while also encouraging users to pay closer attention to the instructions and enter information correctly.

```
Enter your choice:
 E for Encryption
 D for Decryption
 B for Brute Force Attack
 M for Memory
 0 for Exit
0
Exiting the program.

Process finished with exit code 0
```

How to use it:

You start by selecting an option from list provided, then following all other on-screen instructions (provided that there are further instructions for the selected option).



How someone unauthorized is prevented from seeing your message:

The system prevents unauthorized access to your message by requiring a shift value paired with the decrypted message. You need both to access the message intended to be kept private.

Difficulties found:

With the approach we took, there were no major difficulties encountered during the development or use of the program. Understanding the fundamentals of how Python’s built-in functions handle different data types made a significant difference. With this knowledge, we were able to develop a clearer understanding of how Python can be used to manipulate data and achieve the desired results. This demonstrated that once the basic concepts are understood, ideas can be developed into practical solutions through the effective use of Python.

How to make it more user friendly

We have made it as user friendly as possible already. By adding a GUI it would make it more visually appealing, however the functionality would remain the same.